Process Step	**Raw Water Source**
Purpose	Intake for industrial usage
Typical Technologies	Freshwater (municipal), surface water (river), rain, brackish/salt
Notes / Alternatives	Source quality dictates pre-treatment requirements

Pre-treatment	**Primary Treatment**
Protect downstream systems	Solid & settleable removal
Screens, Grit Removal, Coagulation/Flocculation, Sand Filtration, UF	Sedimentation, DAF (Dissolved Air Flotation)
Removes solids to prevent fouling downstream	Often basic step prior to biological or membrane processes
Mechanical Screening & Grit Removal — low cost, protects downstream equipment. **Coagulation/Flocculation + Sand/Media Filters** — common for high TSS influents. **Ultrafiltration (UF)** — improved solids removal and pathogen reduction, ideal before RO .	**Common to Advanced:** **Activated Sludge (CAS)** — baseline for many industries. **MBBR / Biofilm** — compact and robust biologically driven process . **MBR (Membrane Bioreactor)** — high effluent quality; ready for reuse or RO feed . **Anaerobic-MBR (AnMBR)** — wastewater energy recovery (biogas) and low sludge yields . **Specific / Emerging:** **Constructed Wetlands** — decentralized and low-energy (less used in heavy industry) . **Advanced Oxidation (AOP)** — especially for recalcitrant compounds .

Secondary/Main Treatment	Tertiary/Post Treatment	Concentrate / Reject Management
Organic & nutrient removal	Polishing & regulatory compliance	Brine-handling & recovery
CAS (Activated Sludge), MBBR, SBR, MBR (membrane bioreactor), Anaerobic Treatment	UF/NF/RO, Advanced Oxidation (AOP), Activated Carbon, UV, Ion Exchange	Brine concentration (membrane or thermal), Electrodialysis, Membrane Distillation (e.g., Aquaver), MEE/Thermal EV, Crystallization
MBR integrates biological and membrane filtration for compact, high quality effluent	For reuse or stringent discharge requirements	Key for ZLD/MLD — see below
	NF / RO — central for water reuse and desalination.	**Commercial Solutions:**
	Activated Carbon / UV — for micropollutants and disinfection.	**RO Brine Concentration + Thermal Evaporation (MVR/MEE) + Crystallizer** — industry standard ZLD architecture .
	Ion Exchange / Electrodialysis — selective ion removal and partial concentrate control.	**BrineX / Pre-Concentration Membrane Stages** — reduce thermal loads .
		Membrane Distillation / Novel Physicochemical Methods — emerging and research stage .

Zero Liquid Discharge (ZLD)

Circular water use — no discharge

Multi-stage RO + Brine Concentrator + Thermal Evaporators + Crystallizers

Highest water recovery, hig